A Low-Cost EEG and Head-Motion Seizure Alarm: Code-Faithful Evaluation and an Embedded Multimodal Detector

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***Abstract*—Home seizure alarms built from low-cost amplifiers and microcontrollers are rarely tested against annotated seizures. We describe an open-source wearable combining single-channel EEG (EXG BioAmp Pill) and an MPU9250 head-motion sensor on an ESP32, and evaluate its firmware by *code-faithful replay* of public recordings through the device's signal chain. On CHB-MIT (23 subjects, exact FP1–FP2 montage) the shipped detectors were near chance, while a delayed-median ratio rule on the firmware's own EEG deviation (DEV-BR) detected 52.1% of seizures at 0.97 false alarms per hour (FA/h) in held-out subjects, unaffected by gain, clipping and sampling errors. On SeizeIT2 (125 patients, 11,626 h, 883 seizures), requiring EEG *and* head motion to rise together detected 76.4% (95% CI 63–88%) of focal-to-bilateral tonic–clonic seizures at 0.48 false alarms per day, versus 30.9% for EEG and 23.6% for motion alone. The detectors need under 7 KB of RAM and the v5 firmware reproduces the evaluation exactly. Non-convulsive focal seizures remain hard to detect, and the device has not been tested on people.**

***Index Terms*—seizure detection, wearable EEG, accelerometer, multimodal sensing, consumer health, ESP32**

I. Introduction

Unwitnessed nocturnal tonic–clonic seizures are the main risk factor for sudden death in epilepsy [1]. Clinically endorsed wearables detect them from movement or autonomic signals at 90–96% sensitivity and 0.2–0.67 false alarms per day [2], [3]. Open front-ends [4] and Wi-Fi microcontrollers [5] make an EEG alarm cheap, but such devices ship hand-tuned thresholds whose code is rarely tested as written. Baseline normalization is known to help EEG detection [6], a learned EEG–accelerometry model has been shown on SeizeIT2 [7], and event-based scoring is standard [8]. We contribute: (1) a built EEG + MPU9250 prototype; (2) code-faithful replay of its shipped code on two public datasets with subject-level validation and empirical chance levels; (3) embedded detectors that compare EEG and motion with the wearer's own delayed median, fused with *or* (general mode) and *and* (convulsive mode), in under 7 KB of RAM and verified bit-exact.

II. System

One forehead channel (FP1–FP2) is amplified (gain ≈1000, 0.5–40 Hz, 3.3 V supply; vendor documents 5 V), buffered by an LM358 (output ceiling ≈1.8 V at 3.3 V [9]) and sampled by the ESP32 at ≈200 Hz; an MPU9250 [10] (±4 g, ±500 °/s, 10-Hz low-pass, 25 Hz) is read over I^{2}C (Fig. 1). Table I lists what is built, simulated and measured. Alarms drive an RGB LED, a GPIO pulse and a Wi-Fi REST API for a phone app. The MPU9250 and front-end have not been bench-characterized. The shipped (v4) firmware thresholds the deviation $d[n]{=}|x[n]-\lfloor b[n]\rfloor|$ from a 0.5-s baseline at $\mu{+}4\sigma$ of a 1-s-memory running estimate; a Python analyzer votes on line-length, delta/alpha ratio and 30–45-Hz energy.

TABLE I
Implementation status and source of each specification.

Item	Specification	Status / source
EEG front-end	Ag/AgCl FP1+/FP2−; BioAmp Pill (gain ≈1000, 0.5–40 Hz, 3.3 V); LM358 follower; RC 159 Hz	built; nominal values; vendor documents 5-V use [4]
Motion sensor	MPU9250, ±4 g, ±500 °/s, 10-Hz DLPF, 25 Hz, I^{2}C GPIO21/22 [10]	built; not bench-characterized
ADC, sampling	12-bit, default attenuation, `delay(5)` loop	firmware v5 reports measured rate
Outputs	RGB LED, 500-ms GPIO pulse, Wi-Fi AP REST API, phone app	firmware; app not evaluated
Detectors	v4: state machine + 3-feature analyzer; v5: DEV-BR, MOV-BR, co-activation	v4 replay-evaluated; v5 compiled (792 KB flash), bit-exact offline
Bench / human data	noise, battery, timing, wear	not measured

III. Method

Data. CHB-MIT [11], [12] gives the exact device montage (FP1–FP2 reconstructed from the bipolar chain; 160 recordings, 194 seizures, 220 h, 23 subjects) but no motion. SeizeIT2 [13] gives 11,626 h from 125 patients with focal epilepsy: behind-the-ear EEG and a 25-Hz accelerometer on the back of the neck (55 focal-to-bilateral tonic–clonic (FBTC), 371 focal motor, 324 focal non-motor and 133 other seizures; 386 of 2850 recordings lack motion). **Replay.** EEG passes a causal model of the nominal front-end (0.5–40 Hz, 200 Hz, gain 1000, 12 bits); the analyzer runs unmodified on a

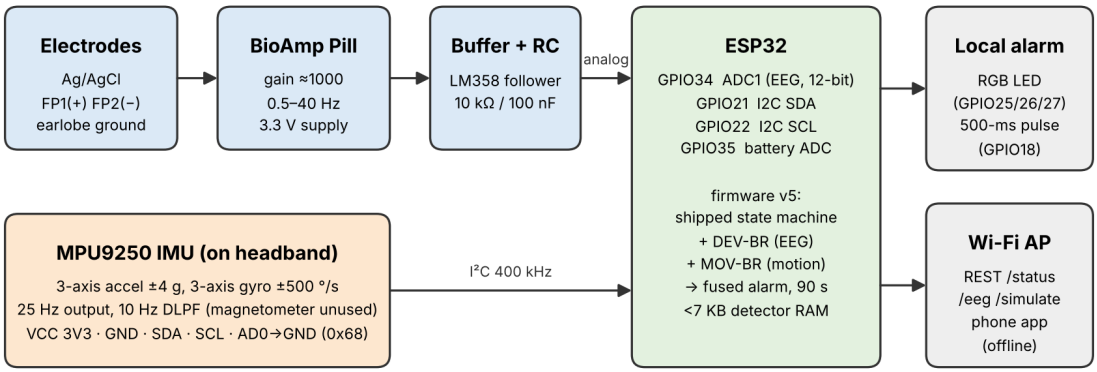


Fig. 1. Prototype (firmware v5): forehead EEG via BioAmp Pill and LM358 to the ESP32 ADC; MPU9250 on the headband over I^{2}C at 25 Hz. Gain and pass-band are nominal, not measured.

simulated serial port and clock, and the firmware loop is ported line by line. **Scoring.** SzCORE [8]: alarms within 90 s merge; detection window [onset−30 s, offset+60 s]. Parameters are chosen by leave-one-subject-out (LOSO) as the most sensitive setting meeting an FA/h target on the other subjects. CIs: subject bootstrap; tests: Wilcoxon (subjects), exact McNemar (seizures); chance: circularly shifted alarms.

Detectors. With $\delta(t)$ the per-second mean of d and $\alpha(t)$ the per-second mean of $\|\mathbf{a} - \mathrm{EMA}_{1\,\mathrm{s}}(\mathbf{a})\|$, each signal is compared with its own median over $[t-329, t-30]$: $\rho_s(t) = \log\big(s(t)/\mathrm{median}\big)$, so gain and sensor units cancel. DEV-BR (EEG) and MOV-BR (motion, optionally gated by 2–7-Hz rhythm) fire when $\rho \geq \log r$ for D s. The general mode fires on EEG *or* motion; the convulsive mode when $\rho_\delta \geq \log r_E$ *and* $\rho_\alpha \geq \log r_M$ for D s. Firmware v5 maps them to the existing BROWNOUT and BLACKOUT states with the modal LOSO parameters (convulsive: r_E=4, r_M=8, D=30 s, chosen in 122/125 folds). It needs under 7 KB of RAM, compiles for the ESP32 (792 KB flash), and its C++ detector reproduced every evaluation alarm (2850 of 2850 SeizeIT2 recordings identical, 11,743 alarms); it has not been timed on the device.

IV. Results

CHB-MIT (device montage). The analyzer's alarm (Table III) detected 5.7% at 0.32 FA/h; the firmware's 75.3% at 18.29 FA/h was largely explained by its alarm rate (chance 56.6%). DEV-BR detected 52.1% (CI 36–74%) at 0.97 FA/h, better than the analyzer in 21 of 23 subjects and worse in none (p=6×10^{-5}); unnormalized thresholds reached 32.5%, and learned classifiers did no better. With parameters frozen it held 50.0%–52.1% across gains of 500–2000, LM358 clipping, a 3.1-V ADC range and a 6-ms loop, while the firmware ranged 7.36–24.37 FA/h (simulation).

SeizeIT2. The shipped firmware detected 57.0% at 7.01 FA/h (chance 29.9%); at that alarm rate EEG-or-motion detected 68.7% (53 patients better, 21 worse, p=4×10^{-5}). At ≤1 FA/h, DEV-BR detected 33.1% and the general mode 31.1%, with large differences by class (Table II, Fig. 3a):

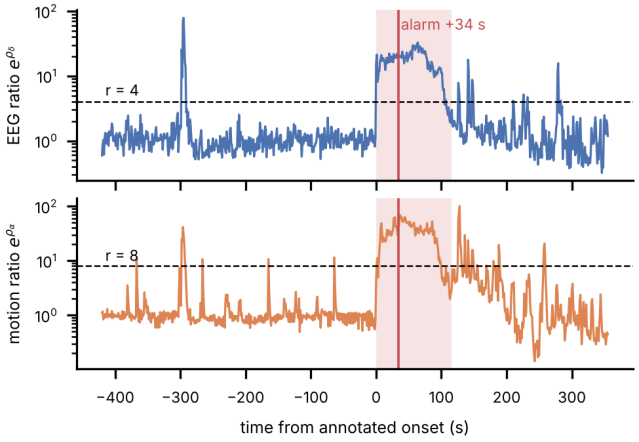


Fig. 2. Convulsive-mode alarm on an FBTC seizure (shaded) in SeizeIT2 patient 115, arising from sleep. Brief EEG artifacts and isolated movements cross one threshold but not both; the seizure raises both ratios and the alarm fires after the 30-s persistence. Shown: the detected sleep FBTC seizure with the shortest latency and no false alarm within 30 min (illustrative, not typical).

96.4% of FBTC but 15.7% of focal non-motor seizures. In convulsive mode, co-activation detected 42/55 FBTC seizures from 37 patients (76.4%, CI 63–88%) at 0.48 false alarms per day, 20 of 25 from sleep (Fig. 2), versus 30.9% (EEG only; McNemar p=4×10^{-7}) and 23.6% (motion only; p=3×10^{-8}) at the same rate; median latency 72 s from the focal onset.

V. Discussion and Conclusion

Replaying the shipped code showed that the prototype's original alarms mostly fire at random, and one rule—compare each signal with the wearer's own delayed median—fixes the EEG detector and transfers to the MPU9250 without calibration. Adding head motion changed the result for the most dangerous seizures: neither modality alone reached a usable alarm rate, but their co-activation detected about three in four convulsive seizures inside the false-alarm range of phase-3 wearables, in under 7 KB of RAM. Performance remains below clinical devices (90–96% [2]) and a learned model on the

TABLE II

SEIZEIT2, HELD-OUT PATIENTS (LOSO) EXCEPT THE SHIPPED FIRMWARE. CLASS COLUMNS: SHARE OF SEIZURES OF THAT CLASS DETECTED; "SLEEP": SEIZURES ARISING FROM SLEEP.

Detector	Detected	Sensitivity [95% CI]	FA/h	FBTC	Focal motor	Focal non-motor	Sleep	Chance
Shipped firmware (BLACKOUT)	503/883	57.0% [47–65]	7.01	87.3%	55.3%	54.6%	64.0%	29.9%
Same false-alarm rate as the shipped firmware								
EEG: DEV-BR	558/883	63.2% [54–72]	5.73	94.5%	74.7%	47.8%	70.8%	–
EEG or motion	607/883	68.7% [57–79]	6.28	98.2%	80.1%	53.4%	83.1%	–
General mode, ≤1 FA/h target								
EEG: DEV-BR	292/883	33.1% [23–44]	0.93	92.7%	41.8%	16.0%	32.2%	4.7%
Motion: MOV-BR	250/883	28.3% [22–36]	1.00	90.9%	39.1%	10.8%	28.3%	4.3%
EEG or motion	275/883	31.1% [22–42]	0.97	96.4%	36.7%	15.7%	32.2%	4.7%
Convulsive mode, chosen for FBTC at ≤0.02 FA/h (≈0.5/day)								
EEG only	21/883	2.4% [1–4]	0.020	30.9%	0.5%	0.0%	3.8%	0.2%
Motion only	14/883	1.6% [1–3]	0.019	23.6%	0.0%	0.3%	1.4%	0.1%
EEG AND motion	75/883	8.5% [4–15]	0.020	76.4%	5.4%	1.2%	11.2%	0.4%

TABLE III

CHB-MIT, FP1–FP2 (23 SUBJECTS, 194 SEIZURES, 220 H; NOMINAL FRONT-END). BRACKETS: SUBJECT-BOOTSTRAP 95% CI. "MACRO": MEAN OF PER-SUBJECT SENSITIVITIES. "CHANCE": SENSITIVITY OF RANDOMLY CIRCULARLY SHIFTED ALARMS. BR ROWS: HELD-OUT SUBJECTS (LOSO).

Detector	Detected	Sensitivity [95% CI]	Macro	FA/h	Latency (s)	Chance
Analyzer, BLACKOUT (shipped)	11/194	5.7% [2–12]	10.0%	0.32	26.3	1.6%
Analyzer, any alarm (shipped)	41/194	21.1% [14–29]	25.4%	2.82	28.4	12.8%
Firmware, BLACKOUT (shipped)	146/194	75.3% [60–94]	87.6%	18.29	14.5	56.6%
LL-BR, ≤1 FA/h	49/194	25.3% [15–42]	40.1%	1.04	32.0	4.8%
FUSE-BR, ≤1 FA/h	78/194	40.2% [24–64]	56.0%	0.96	21.5	–
DEV-BR, ≤1 FA/h	101/194	52.1% [36–74]	68.6%	0.97	22.0	6.5%
DEV-BR, ≤0.5 FA/h	81/194	41.8% [26–65]	59.5%	0.46	23.0	3.9%
DEV-BR, ≤0.2 FA/h	55/194	28.4% [15–49]	46.1%	0.16	29.0	–

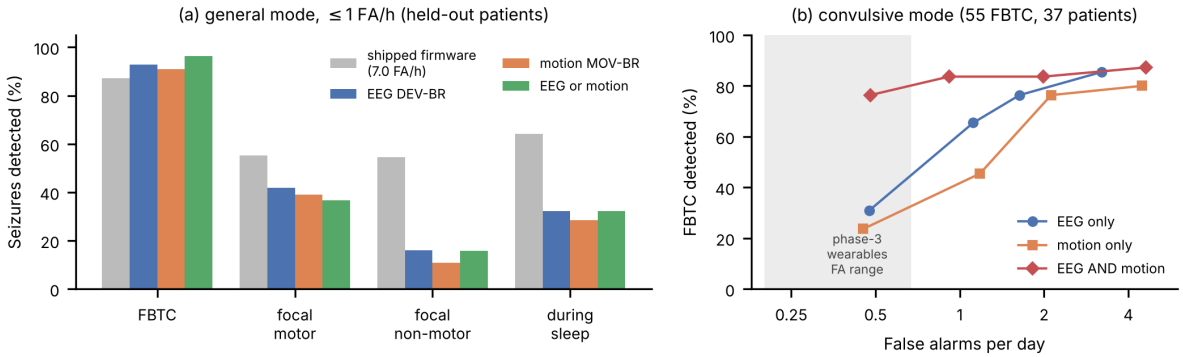


Fig. 3. SeizeIT2, held-out patients. (a) Seizures detected by class at ≤1 FA/h (shipped firmware at its own 7.0 FA/h). (b) Convulsive mode: FBTC sensitivity versus false alarms per day for LOSO targets of 0.02–0.2 FA/h; shaded, the phase-3 wearable range [2].

same data (95.5% at 1.1/day [7]), and non-convulsive seizures are rarely caught. Limitations: results come from software replay; SeizeIT2's behind-the-ear EEG and neck-worn 25-Hz sensor differ from our forehead headband; FBTC results rest on 55 seizures; parameters are modal LOSO choices; no bench, timing or human data are reported. Next steps are bench characterization, overnight headband recordings and an ethics-approved study. Hardware, firmware, replay code and detectors are open source [14].

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